

Homework 3

NYC Taxi Analytics with Pyspark

1. EC 2 Setup

The analysis was performed on an Amazon EC2 instance (t2.large, 80 GB gp3 storage) running Amazon Linux 2023 in the us-east-1 region. The instance was accessed via SSH using a .pem key pair. The code was then run on Jupyter Notebook, which I accessed from a local machine via SSH port forwarding (port 8888). In each run of the lab, I refreshed AWS from the AWS port. The analysis used Python version 3.11.6 and PySpark version 4.1.1.

2. Dataset Month Used

In the analysis, I used yellow and green taxi data from January 2026. During this month, there were 3,724,889 rows of Yellow Taxi data, and 40,272 of Green taxi data.

3. Data Cleaning Assumptions

Before cleaning, the two datasets required schema standardization to be merged. The yellow taxi dataset used `tpep_pickup_datetime` and `tpep_dropoff_datetime` for timestamps, while the green taxi dataset used `lpep_pickup_datetime` and `lpep_dropoff_datetime`. Both columns were renamed to a unified `pickup_datetime` and `dropoff_datetime`. A `taxi_type` column was also added to each dataset (labeled "yellow" or "green") to distinguish the source after merging. Finally, only columns shared between both datasets were retained. The two DataFrames were combined using a union operation. Once the data was combined, I wrote a series of functions to clean the data. They removed a total of 208,124 rows, and the logic used is as follows:

- **Null timestamps:** Rows with missing pickup or dropoff datetimes cannot be meaningfully analyzed. Filtered using `.isNotNull()` on both columns. *Rows removed: 0*
- **Zero or negative trip distance:** Trips with a distance of zero or less likely represent cancelled or erroneous entries. Filtered using `trip_distance > 0`. *Rows removed: 127,028*
- **Negative fare amount:** Negative fares indicate data entry errors. Filtered using `fare_amount >= 0`. *Rows removed: 39,582*
- **Negative total amount:** Same reasoning as fare amount; negative totals indicate corrupted records. Filtered using `total_amount >= 0`. *Rows removed: 40,104*
- **Dropoff before pickup:** If dropoff time occurs before or at the pickup time it indicates corrupted timestamp. Filtered using `dropoff_datetime > pickup_datetime`. *Rows removed: 45,099*

- **Trips longer than 24 hours** — Trips exceeding 24 hours are almost certainly data entry errors rather than real trips. Filtered by computing the difference in Unix timestamps and keeping only those $\leq 86,400$ seconds. *Rows removed: 35*

Note: some rows may violate multiple rules, so the individual counts won't add up exactly to the total of 208,124 removed.

4. Answers to Analytics Questions

1. Which taxi type had more trips?

Yellow taxis had significantly more trips than green taxis in January 2026. Yellow took **3,518,128** trips compared to **38,909** for green taxis, meaning yellow taxis made up ~98.9% of all trips in the dataset. This makes sense because Yellow taxis are the dominant taxi type in Manhattan while green taxis are restricted to outer boroughs.

2. What was the average fare by taxi type?

Yellow taxis had a higher average fare of **\$21.08** compared to **\$16.00** for green taxis. This difference is likely explained because yellow taxis operate in more high traffic areas, are more likely to make airport runs, and may be subject to Manhattan congestion fees.

3. What was the average trip distance by taxi type?

Despite having lower average fares, green taxis had a notably higher average trip distance of **12.99** miles compared to **6.76** miles for yellow taxis. Given that NYC charges special fees for Manhattan trips below 96th Street and that outer boroughs tend to be less densely populated with lighter traffic, it makes sense that green taxi trips are less expensive despite covering greater distances.

4. What hour of day had the most pickups?

The peak pickup hour was 6 PM (hour 18) with 232,570 trips. The top hours were all concentrated in the afternoon and evening (2 PM–9 PM), reflecting end-of-workday commuter demand in NYC. The quietest hours were in the early morning between 4–6 AM, with 4am having the fewest trips at just 27,792.

Hour	12:00A M	1:00AM	2:00AM	3:00AM	4:00AM	5:00AM	6:00AM	7:00AM	8:00AM	9:00AM	10:00A M	11:00A M
Trip #	109,013	75,623	52,470	37,491	27,792	32,070	62,649	112,027	146,020	154,087	158,111	167,923

Hour	12:00P M	1:00PM	2:00PM	3:00PM	4:00PM	5:00PM	6:00PM	7:00PM	8:00PM	9:00PM	10:00P M	11:00P M
Trip #	183,146	190,970	202,317	216,305	206,485	224,277	232,570	211,118	208,072	207,428	190,251	148,822

5. What percentage of trips were under 2 miles?

51.95% of all trips (**1,847,954 out of 3,557,037**) were under 2 miles, meaning more than half of all NYC taxi trips in January 2026 were very short. This reflects the dense urban environment of New York City where many riders use taxis for quick neighborhood trips, particularly during the cold winter month of January.

8. Predict the fare amount using non-fare predictor columns.

A Random Forest Regressor was trained to predict **fare_amount** using 11 non-fare predictor columns. Rows with any null values in the feature columns were dropped before training. Features were assembled into a single vector using **VectorAssembler**, and the data was split 80/20 into training (2,045,084 rows) and testing (511,300 rows) sets using a fixed seed of 42. The model was configured with 50 trees and a max depth of 5.

RMSE Results:

- Training RMSE: 6.8108
- Testing RMSE: 6.7353

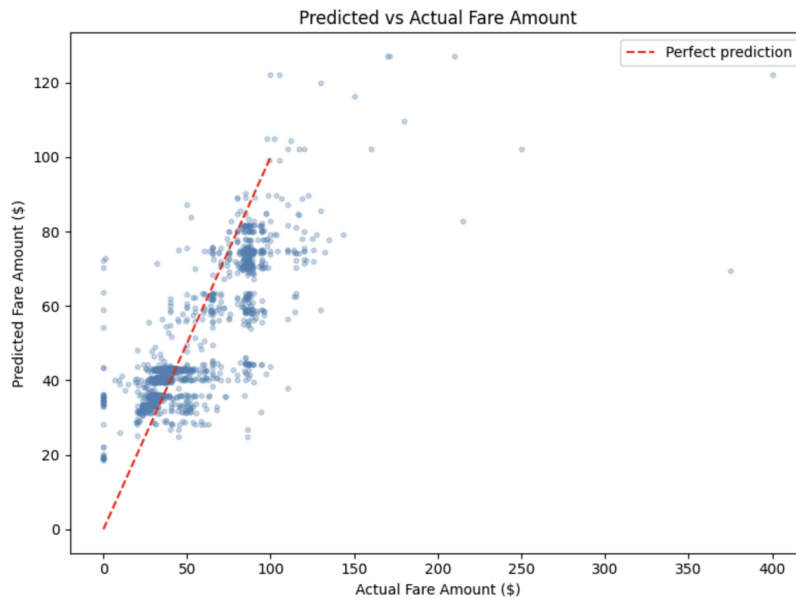
The training and testing RMSE are nearly identical, indicating the model generalizes well with no significant overfitting.

Feature Importance

Feature Importances: Trip distance was by far the most important predictor at 46.9%, which makes intuitive sense as longer trips directly drive higher fares. Tip amount (19.6%) and tolls amount (17.9%) were the next most important, likely because they are correlated with longer or more expensive trips. Fixed surcharges like MTA tax, congestion surcharge, and improvement surcharge had moderate importance. Pickup and dropoff location IDs contributed modestly, capturing geographic pricing differences. Passenger count had negligible predictive power at just 0.005%, confirming that the number of passengers has almost no effect on fare amount in NYC taxis.

Feature	Importance
trip_distance	0.4686
tip_amount	0.1964
tolls_amount	0.1787
mta_tax	0.0433
congestion_surcharge	0.0428
PULocationID	0.0392
DOLocationID	0.0165
improvement_surcharge	0.0076
extra	0.0039
cbd_congestion_fee	0.0029
passenger_count	0.0001

Predicted vs Actual Plot:



The scatter plot shows predictions tracking reasonably well with actual fares in the \$20–\$100 range, with more spread at higher fare values. The model tends to underpredict very high fares (above \$150), which is expected given those are rare outliers that are harder to capture with a depth-5 tree.

S3 Paths:

- `s3://de300-hw3-westra/nyc-taxi-assignment/trips_by_type.csv`
- `s3://de300-hw3-westra/nyc-taxi-assignment/avg_fare_by_type.csv`
- `s3://de300-hw3-westra/nyc-taxi-assignment/pickups_by_hour.csv`
- `s3://de300-hw3-westra/nyc-taxi-assignment/predicted_vs_actual.png`

AI Usage: Claude (Anthropic) was used to help debug PySpark setup, fix credential errors, and structure the notebook